

Courses Taught

Below is a list of courses and outreach lectures I have delivered, updated to November 2025.

As a Forensic Inspector of the Forensic Department of the Italian Police:

- **Training Courses for Forensic Trainees.** Prepared and delivered 19 hours of lectures and hands-on training sessions on Electron Microscopy, Energy-Dispersive Spectroscopy, and standard forensic procedures for Gunshot Residue analysis, offered to the trainees of the Forensic Inspector course. (*Caserta, Nov. 2025*)

As a (Master's and) Ph.D. student at SNS:

- **Physics School in Pisa.** Prepared and delivered 28 hours of lectures on topics in Classical and Modern Physics, and led related problem-solving sessions on Fluid Dynamics, Celestial Mechanics, Special Relativity & Introduction to Quantum Mechanics, as well as Problem-Solving Techniques and Physical Analogies. As a member of the organizing committee, I also examined and graded approximately 200 applicants per year through both written and oral exams. (*Pisa, 2019–2025*)
- **“University Orientation Courses”** (*Third Mission* outreach). Designed and taught 12 hours of interactive lessons on Problem-Solving Techniques and Everyday Physics, aimed at 260 selected high school students. (*Three locations across Italy, June–July 2024*)
- **Classical Mechanics and Thermodynamics.** Served as a teaching assistant, delivering 30 hours of problem-solving sessions for first-year university students. (*Pisa, Fall 2020 – Spring 2021*)

As a Physics Olympiad trainer:

- **“One Step Away from IPhO”.** Prepared and delivered 32 hours of advanced lectures on Celestial Mechanics, Electromagnetism, and Thermodynamics, designed for the top 15 Italian students preparing for the International Physics Olympiad (IPhO). Together with two other instructors, I selected the national team members participating in IPhO and EuPhO through theoretical and experimental examinations. (*Trieste, May 2019/23/24/25*)
- **Physics Olympiad Summer School.** Led 16 hours of problem-solving sessions on Orbital Mechanics, Estimation Problems, Dimensional Analysis, and Fluid Dynamics, addressed to 20 selected students competing in the Physics Olympiad. (*Sigillo, August 2019/24/25*)
- **Training Camp for the Physics Olympiad.** Prepared and gave 6 lectures on Electrostatics, Problem-Solving Techniques, and Fluid Dynamics for 20 selected students competing in the national finals of the Physics Olympiad. Additionally, I prepared and graded the final theoretical competition each year. (*Catania, March 2023/24/25*)

Beyond teaching activities, I also design original physics problems and provide evaluation rubrics for the *Team Physics Competition* and the *Italian Physics Olympiad*, where I serve as both an organizing and grading committee member.